

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 7

10 Further Human–Computer Interaction (afb21)

In future, many households will have a limited energy budget. Diverse household members are likely to have different energy requirements that must be negotiated within that fixed overall budget.

This question relates to the design of a shared display that can be used by household members to monitor and discuss their respective energy usage. The display will be primarily graphical, rather than using text or numbers.

design

- (a) Make a sketch of the display layout, which addresses the scenario above, and can be used to answer the remainder of the question. Use arrows to label the four elements referred to in the next part. [2 marks]

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*Answer:* Full marks will be awarded to any sketch that addresses the problem and includes four labelled arrows.

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visual vocabulary

- (b) Name *four* different types of graphical mark or property that have been used as the basic elements of the overall design. [2 marks]

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*Answer:* Graphical marks include: linear ordering, coloured regions, photographs, lines, star symbols

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representational  
correspondence

- (c) For *each* of these types of graphical mark, explain:
- (i) What it looks like (its appearance).
- (ii) What it represents (its meaning).
- (iii) The nature of the correspondence between its appearance and meaning in your design.

[12 marks]

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*Answer:* (i) Linear ordering is the left-to-right placement of graphical elements across the screen. (ii) In my design, this represents the time at which energy is being used. (iii) The correspondence is between linear scale of time measurement and distance on the representational surface, and is also consistent with the traditional convention of a time axis being oriented horizontally with earlier times toward the left.

(i) The coloured regions are used to fill an area that corresponds to the amount of energy used each day. (ii) The colour is used to represent the link between usage by each household member on different days, by making the area for that person the same colour each time. (iii) The colour choices are arbitrary, chosen only so that the different colours can be perceptually distinguished (ideally within a colour space that will maintain colour distinction for anyone who is red-green colourblind). Choice of colours could also take into account aesthetic preference, for example offering a choice of palettes that reflect alternative taste in domestic decorating.

(i) The photographs are thumbnail portraits within a round frame, with a drop shadow to make them appear to float above the coloured region. (ii) Each photo represents one member of

the household, and is used to indicate that the coloured region it floats above relates to energy usage for that person. (iii) The correspondence is physical resemblance (within conventions of photographic portraiture) between the person and their photograph.

(i) The lines extend horizontally across the display. (ii) They represent the target energy use, on a vertical scale. (iii) The correspondence is geometric, showing by continuity that the target on every day is the same.

(i) The symbols are abstract five-point stars, coloured gold. (ii) A star will be added within the region for each day that the energy target has been met. (iii) The correspondence is an iconic convention, in which star symbols are often used in a Western context to indicate achievement of a goal. If the design was to be used in a non-Western context, it might be a good idea to check whether interpretation of the star symbol varies there. However, the star convention is relatively widespread in social media, and likely to be familiar from that context, even if the level of motivation from award of symbolic stars differs in different places.

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qualitative  
methods

- (d) Describe how qualitative empirical data could be collected and analysed to evaluate and improve this design concept. [4 marks]

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*Answer:* Empirical data is collected by observing phenomena in the world. The design as described above is dependent on a number of cultural factors, and will be interpreted in the context of the household. The intended outcome of the design intervention is enhancement of dialog among the household members, which could be evaluated by collecting examples of dialog in a household where a prototype version of the system was deployed. This could either be done ethnographically, by an observer embedded in the household, or alternatively by interviewing members of the household in context, either individually or in focus group style. A first level of evaluation would simply transcribe and categorise any negative or critical comments about the value or usability of the device. The design concept could be improved by repairing these faults. More sophisticated design assessment could involve critical analysis of the interpretation of the device, analysed using grounded theory to inform unanticipated creative developments. Finally, it would be possible to work collaboratively with household members in a participatory way, collecting their qualitative input to the co-design process.

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